SUSTAINABLE INDUSTRY SPRINT

Type : Group (4 members)

Duration : 3.5 - 4 hours

“Design the Future. Solve for Sustainability.”

EVENT OBJECTIVES

- **Expose students to real industrial sustainability challenges.**
- **Develop structured problem-solving under time constraints.**
- **Improve professional communication and decision-making.**
- **Encourage interdisciplinary collaboration and strengthen presentation.**

⏳ EVENT FLOW & TIMELINE

🟢 **0:00 – 0:20 | Registration & Orientation**

What Happens:

- Team verification
- Distribution of A4 problem analysis sheets
- Rules & evaluation criteria explained
- Judges introduced (Faculty / Alumni / Industry Guest).

🟢 **0:20 – 0:30 | Problem Statement Reveal**

- One problem sheet is revealed to all teams.
- All teams receive identical data sheets.
- No prior preparation allowed.

Teams must work within given:

- Budget constraints
- Timeline limits
- Technical restrictions

🟢 **0:30 – 2:00 | Analysis Phase (90 Minutes)**

Teams must:

✔ Clearly define the problem

✔ Identify root causes

✔ Brainstorm 2–3 solutions

✔ Select the best solution

✔ Estimate cost

✔ Analyze sustainability impact

✔ Assess risks and feasibility

✔ Develop implementation timeline

Participants record their structured thinking on provided A4 sheets.

🟢 **2:00 – 3:00 | Presentation Preparation (60 Minutes)**

Teams prepare a 5–7 slide PPT including:

- Problem Summary
- Key Insight
- Proposed Sustainable Solution
- Cost Breakdown
- Environmental Impact
- Implementation Plan
- Why This Works

🟢 **3:00 – 3:50 | Boardroom Presentation Round**

Each Team Gets:

• 7–8 minutes Presentation

• 5 minutes Q&A by Judges

Professional attire encouraged.

📜 RULES & REGULATIONS

Team Structure

• Exactly 4 members per team

• Interdisciplinary teams encouraged

Event Format Rules

🔹 **Problem Statement**

- Revealed only at event start
- No external prior preparation allowed
- Same dataset provided to all teams

🔹 **Time Limits**

- **90 min –** Analysis
- **60 min –** PPT Preparation
- **7–8 min –** Final Presentation
- **5 min –** Q&A

🔹 **Permitted Resources**

- Laptops allowed
- Internet usage: (Decide – Allowed / Restricted / Not Allowed)

🔹 **Mandatory Solution Components**

Each team must include:

- Clear Problem Definition
- Root Cause Analysis
- Sustainable Solution Proposal
- Cost Estimation
- Environmental Impact Assessment
- Implementation Timeline
- Risk & Feasibility Analysis

📄 SAMPLE PROBLEM STATEMENTS

PROBLEM SHEET 1

EV Battery Waste Optimization – Clean Mobility Startup

🏭 **Company Background**

“VoltRide Mobility” is a mid-sized Indian EV startup manufacturing electric scooters.

They produce 25,000 lithium-ion battery packs per year.

⚠️ **Current Issue**

- 10% of batteries fail quality tests.
- Disposal cost per rejected battery: ₹2,200
- Increasing regulatory pressure on battery waste.
- Customers demanding sustainability transparency.

💰 **Constraints**

- Budget limit: ₹75 lakhs
- Cannot change battery supplier immediately.
- Solution must be implementable within 12 months.

🎯 **Target Outcome**

- Reduce battery waste by at least 50%.
- Lower disposal cost.
- Improve sustainability reputation.

PROBLEM SHEET 2

Sustainable Packaging Redesign – Food Manufacturing Company

🏭 **Company Background**

"FreshBite Foods" produces packaged snacks sold nationwide.

Annual packaging usage:

- 1.5 million plastic packets
- Cost per packet: ₹4.20

⚠️ **Current Issue**

- Plastic waste criticism from consumers.
- Retailers asking for eco-friendly packaging.
- Packaging must remain moisture-resistant.

💰 **Constraints**

- New packaging cost must stay under ₹5 per unit.
- Shelf life must remain 6 months minimum.
- Machines cannot be completely replaced.

🎯 **Target Outcome**

- Reduce plastic usage by at least 60%.
- Maintain product quality.
- Keep overall cost increase under 15%.

PROBLEM SHEET 3

Campus Energy Reduction Strategy – Private Engineering College.

🏫 **Background**

A 3,000-student engineering college spends **:**

- ₹1.2 crore annually on electricity.
- 40% of usage comes from classrooms and labs.

⚠️ Current Issue

- Rising electricity tariff (increase expected 20% next year).
- ACs and lights often left on unnecessarily.
- Administration wants visible sustainability initiative.

💰 Constraints

- Budget: ₹50 lakhs.
- Solar installation possible but limited rooftop space.
- Cannot disrupt academic schedule.

🎯 Target Outcome

- Reduce electricity bill by 25% within 18 months.
- Improve campus sustainability ranking.
- Achieve measurable carbon reduction.

REQUIREMENTS

📄 Materials

- Printed A4 Problem Sheets.
- Analysis Recording Sheets.
- A room with a projector- In order for the participants to explain/showcase their solution.

👩‍💻 Participants

- Must bring laptops.
- Formal or semi-formal attire recommended.

🎖️ Rewards

Prizes for the winning teams.

🏆 1st Prize – Trophy + Certificate.

🥈 2nd Prize – Certificate.

🌿 Sustainability-themed badges/stickers.